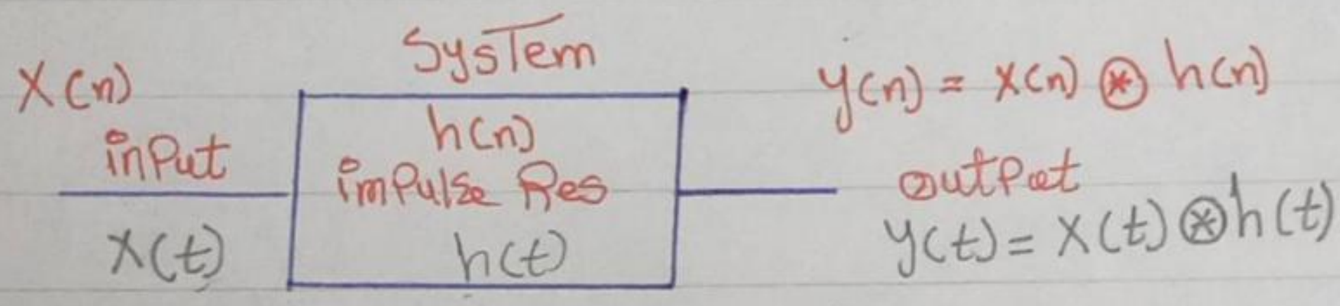
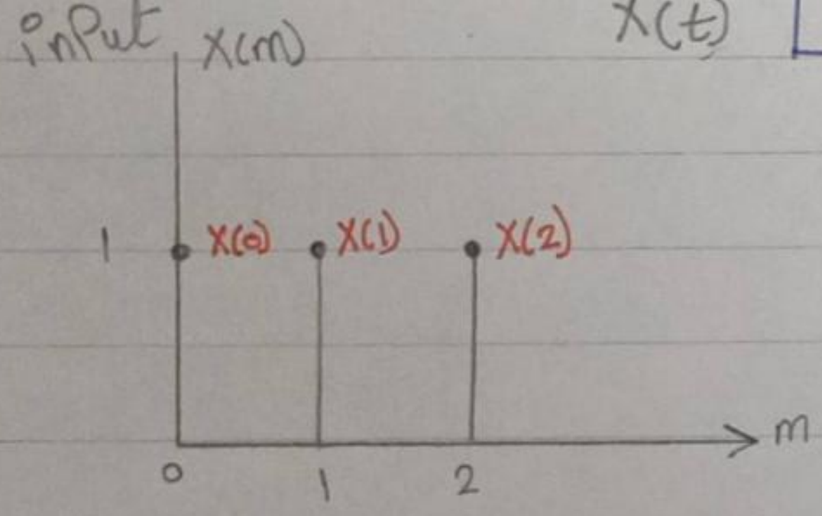
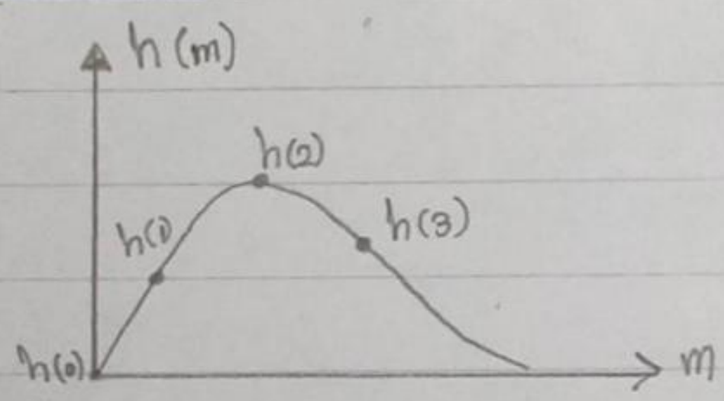
Chapter#3 Convolution

Convolution:-

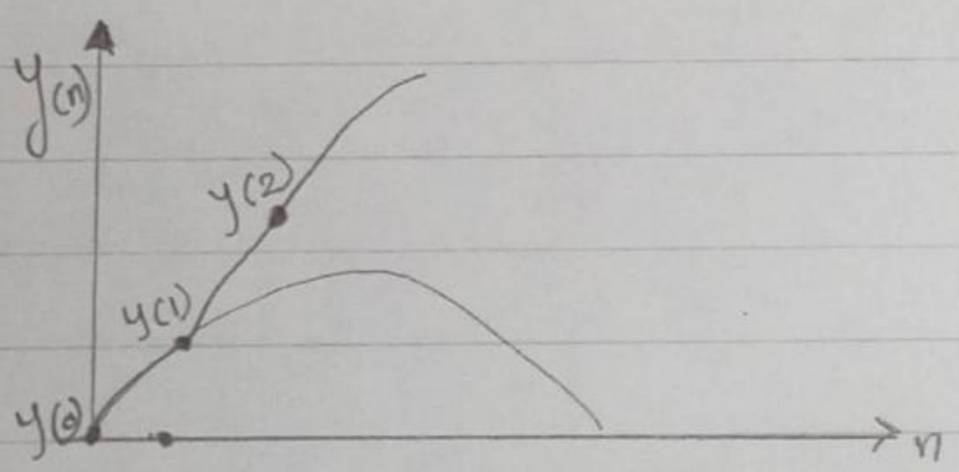
How the input interacts with the system to produce the output.







$y(0) = h(0)\,x(0)$



$y(1) = h(1)\,x(0) + h(0)\,x(1)$

$y(2) = h(2)\,x(0) + h(1)\,x(1) + h(0)\,x(2)$

→ The Convolution in discrete-Signal

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)\,h(n-m) \quad , \quad n = 0 \cdots N-1$$

→ method

① changing the axis from $n \longrightarrow m$
② Fold : mirror one of them $m \longrightarrow -m$
③ shift Right the mirror signal $-m \longrightarrow n-m$
④ summation to get $y(n)$.

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Convolution

→ For The continuous signal → analytically

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

→ For the Discrete-Signal → Direct method → mathmatical

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)\, h(n-m), \quad n = 0 \cdots N-1$$

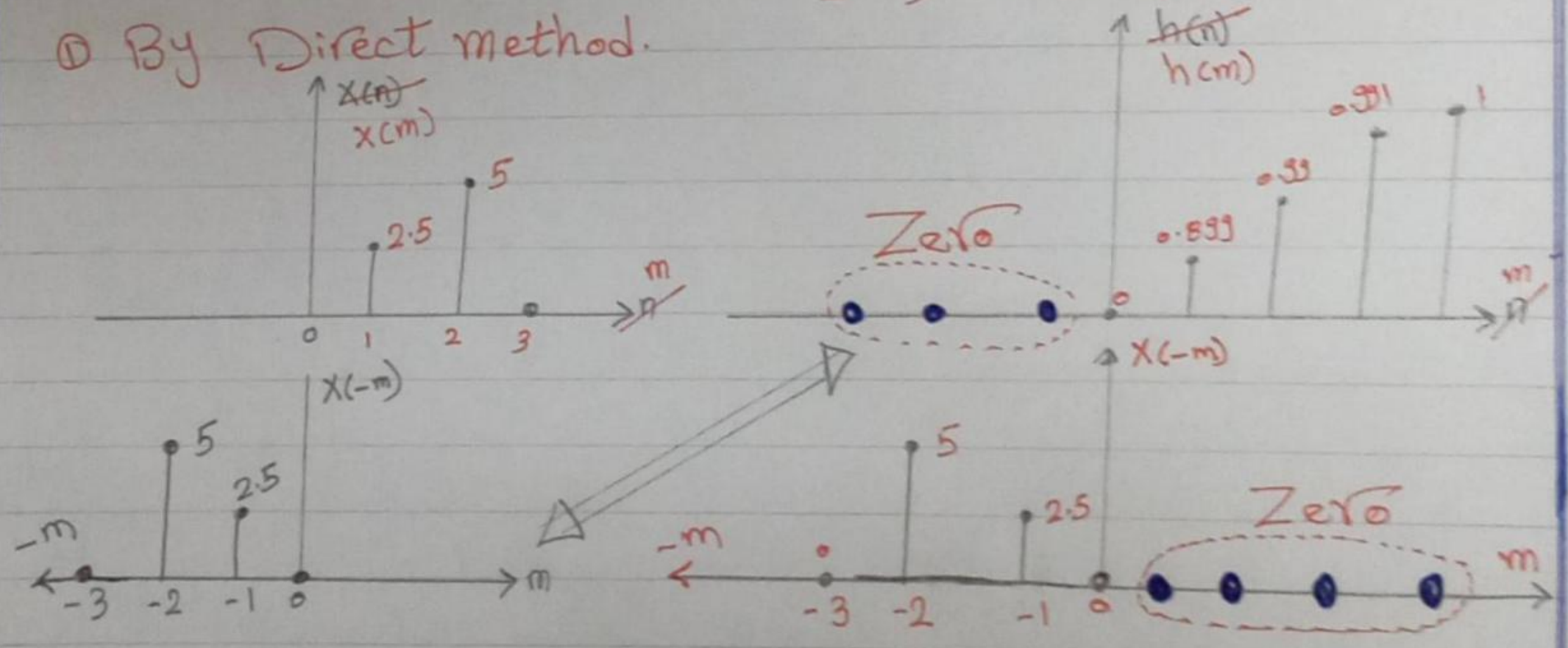
→ Fast Convolution → Convolution theorem

$$y(n) = FD^{-1}\left[X(K)\, H(K)\right]$$

Sheet #3

3.6 Determine the output of an electric system of impulse response Function $\{0, 0.899, 0.99, 0.991, 1\}$ when The input $\{0, 2.5, 5, 0\}$

① By Direct method.



$h(m) = \{0, 0, 0, 0, 0.899, 0.990, 0.991, 1\}$

$X(-m) = \{0, 5, 2.5, 0, 0, 0, 0, 0\}$

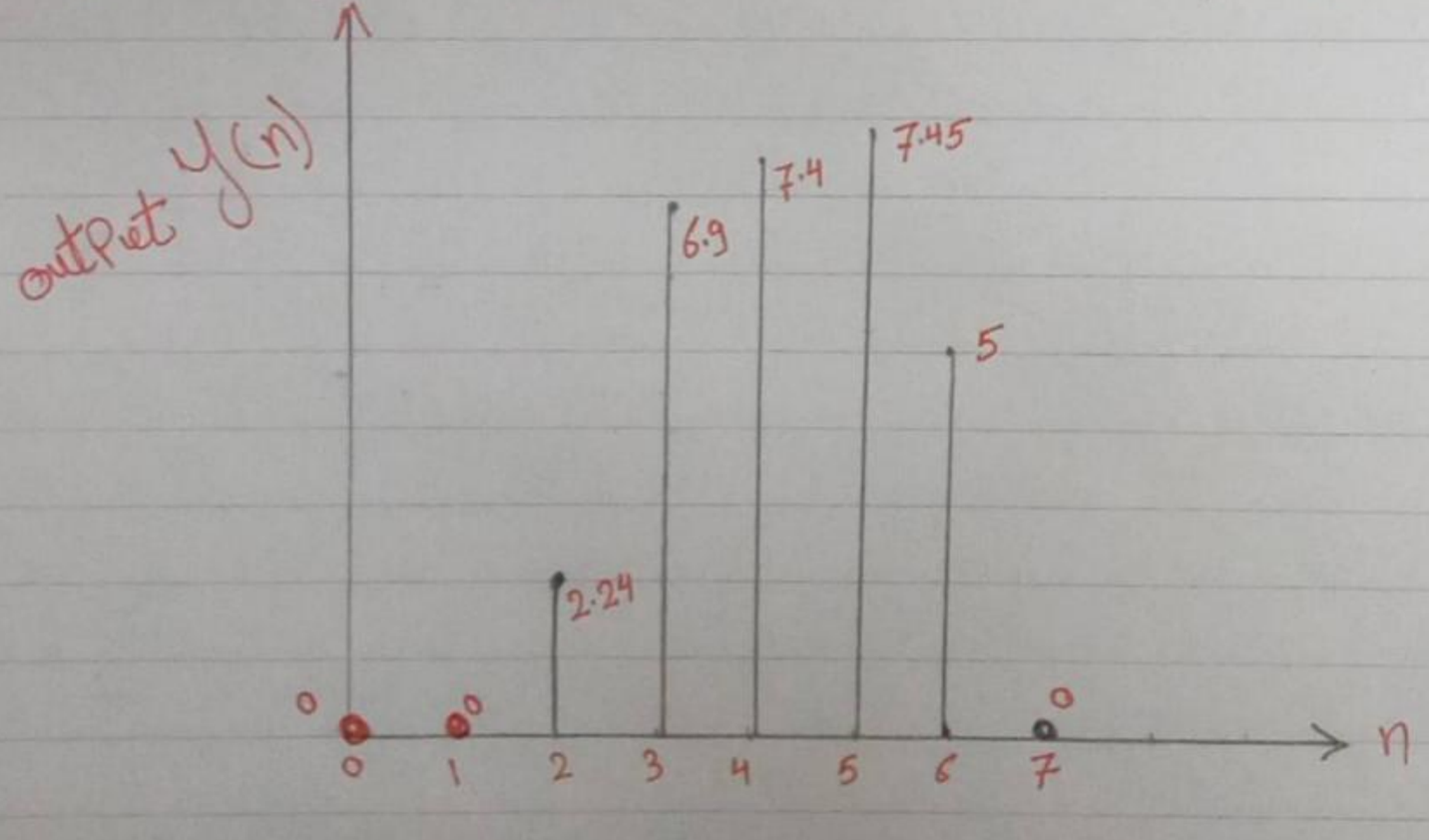
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$$y(n) = \sum_{m=0}^{N-1} h(m)\, x(n-m)$$

shift Right

n	h(m)	0	0	0	0	0.899	0.99	0.991	1	y(n)
0	x(0-m)	0	5	2.5	0	0	0	0	0	0
1	x(1-m)	Zero	0	5	2.5	0	0	0	0	0
2	x(2-m)	Zero	Zero	0	5	2.5	0	0	0	2.24
3	x(3-m)	Zero	Zero	Zero	0	5	2.5	0	0	6.97
4	x(4-m)	0	0	0	0	0	5	2.5	0	7.427
5	x(5-m)	0	0	0	0	0	0	5	2.5	7.455
6	x(6-m)	0	0	0	0	0	0	0	5	5
7	x(7-m)	0	0	0	0	0	0	0	0	0

$$y(n) = \{0, 0, 2.24, 6.97, 7.427, 7.455, 5, 0\}$$



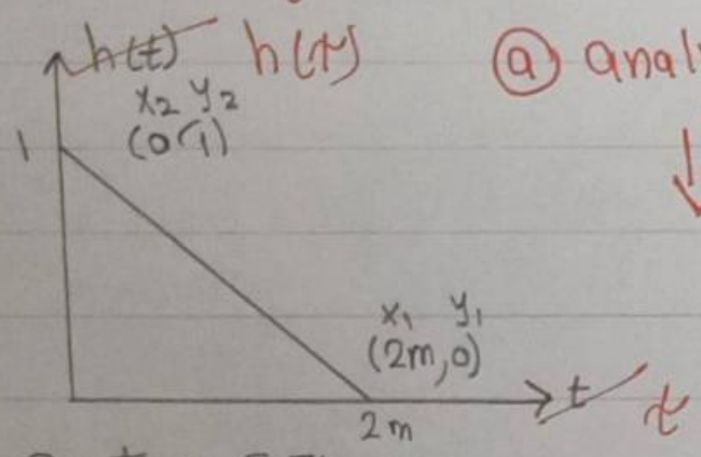
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(3.7) A unit amplitude pulse of width 2m. is applied to a circuit has impulse response in Fig 2. Determine the output.

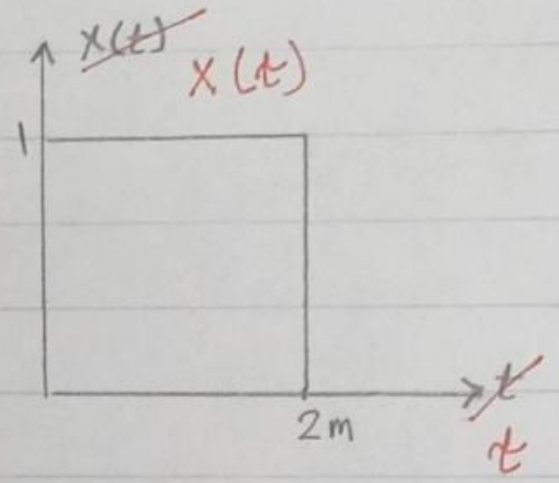
ⓐ Analytically

Ⓑ by sampling h(t) every 0.5m and representing the pulse by an impulse function located at t=0s

Ⓒ Compare your results in Part ⓐ and Ⓑ



ⓐ analytical



* The equation of Fig 1

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-0}{0-2m} = -0.5$$

$$y = mx + c$$

$$1 = 0 + c \quad \boxed{c = 1}$$

$$y = -0.5x + 1$$

$$h(t) = -0.5t + 1$$

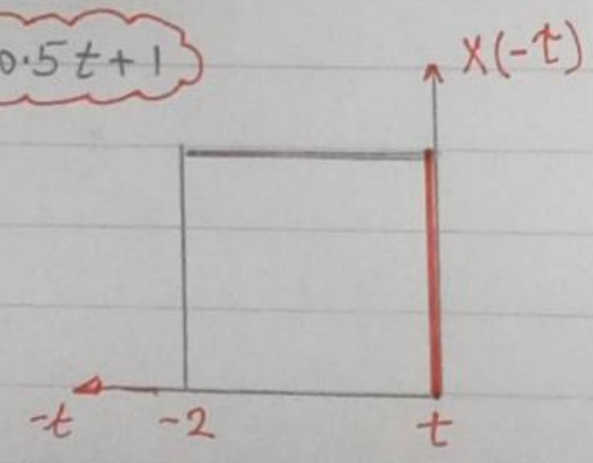
$$x(t) = 1$$

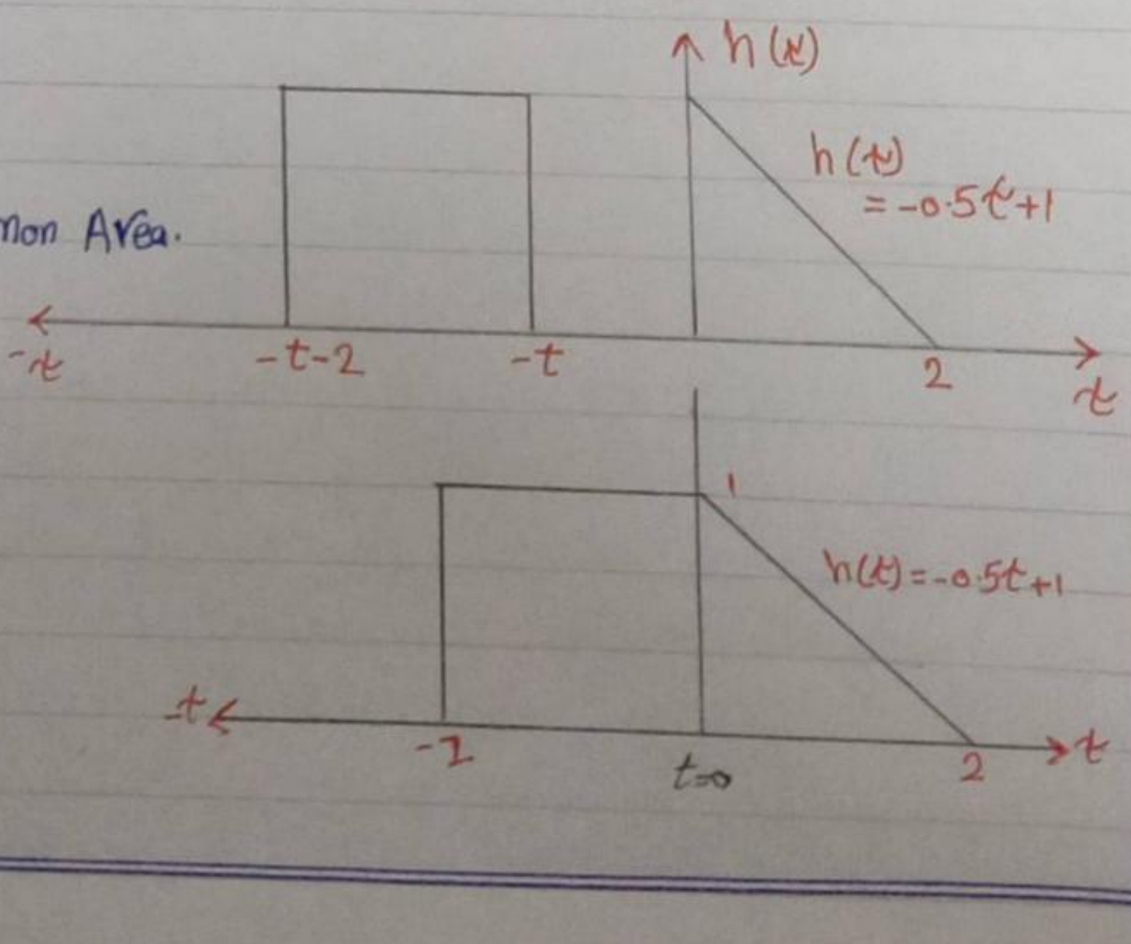
① Changing the axis $t \rightarrow t$

② Fold: mirror one of them $x(-t)$

③ shift right the folding $x(t-t)$

④ integration $\int h(t)\, x(t-t)\, dt$



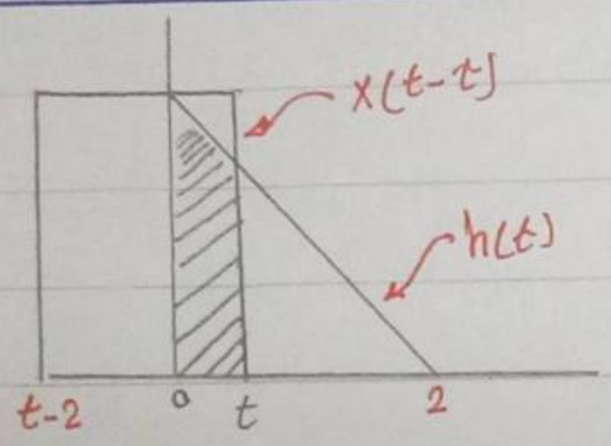


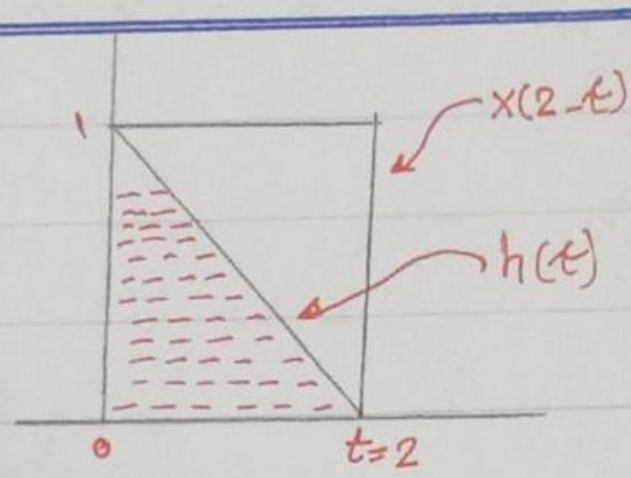
[1] $t \leq 0$ $\quad y(t) = \text{Zero}$

No Common Area.

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② $0 < t \leq 2$

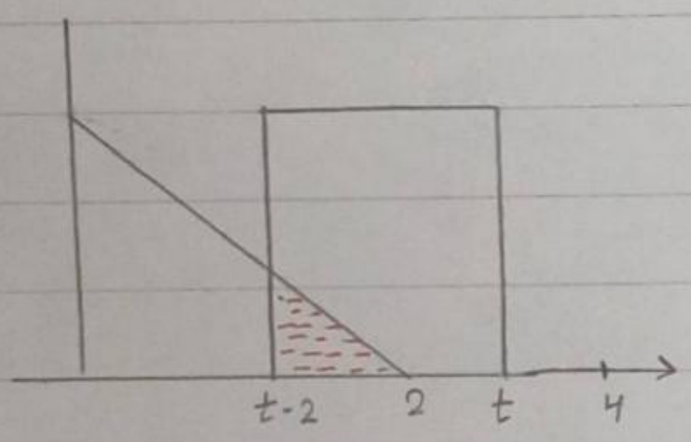


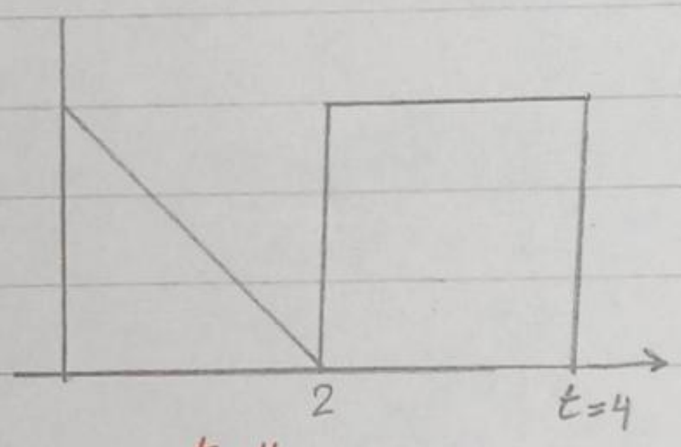


$$y(t) = \int_0^t x(t-t)\,h(t)\,dt = \int_0^t (0.5t+1)\,dt = -0.25t^2 + t\Big|_0^t = t - 0.25t^2$$

at $t=2$ $\quad y(2) = 2 - 0.25\times4 = 1$ & $y(2) = \frac{1}{2}\times2\times1 = 1$

③ $2 < t \leq 4$





$t=4$

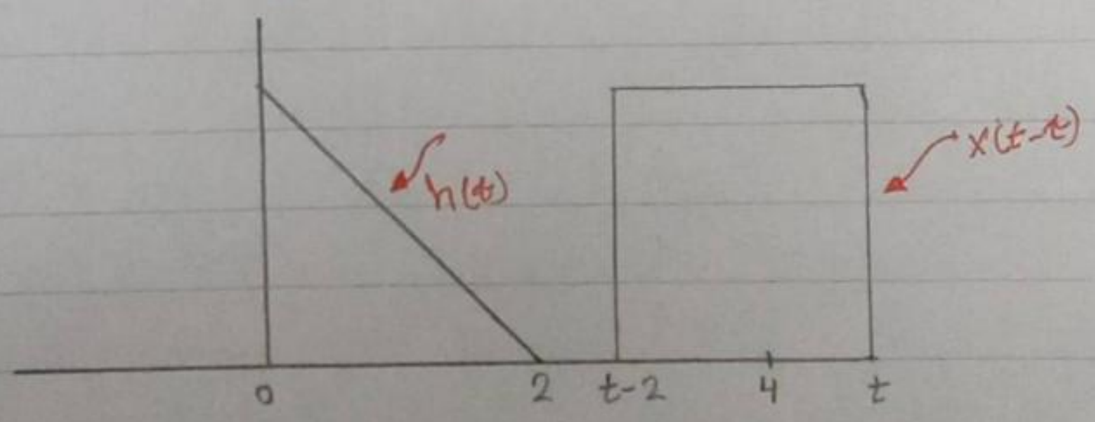
$$y(t) = \int_{t-2}^{2} (-0.5t+1)\,dt = -0.25t^2 + t\Big|_{t-2}^{2}$$

$$= \frac{-1}{4}\cdot4 + 2 - (-0.25(t-2)^2 + t - 2) = 1 - (-0.25t^2 + t - 1 + t - 2)$$

$$= 1 + 0.25t^2 - 2t + 3 = 0.25t^2 - 2t + 4$$

→ at $t=4$ $\quad y(4) = 0.25\cdot16 - 8 + 4 = \text{Zero}$

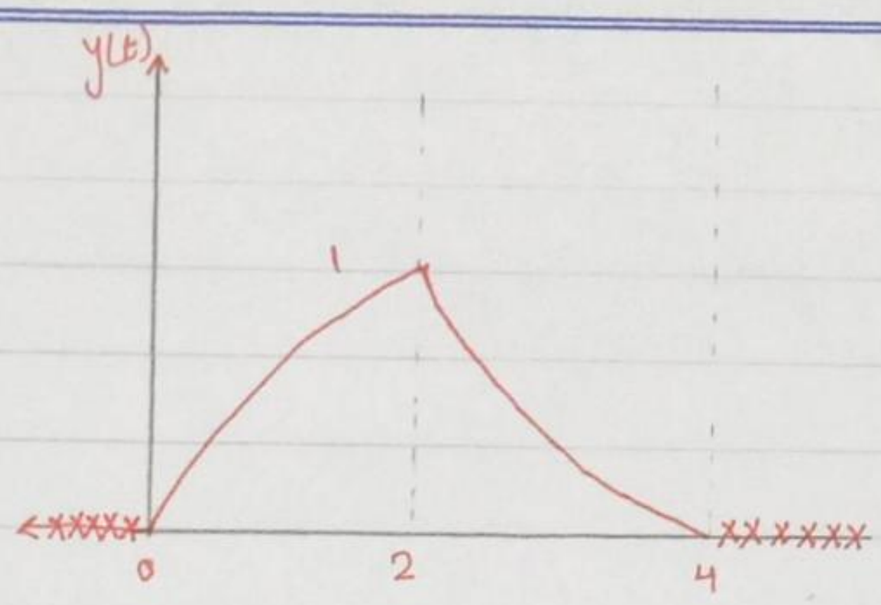
④ $t > 4$



$y(t) = \text{Zero}$ $\quad$ No Area.

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$$y(t) = \begin{cases} 0 & t \le 0 \\ -0.25t^2 + t & 0 < t \le 2 \\ 0.25t^2 - 2t + 4 & 2 < t \le 4 \\ 0 & t > 4 \end{cases}$$



(B) by Sampling $h(t)$, $x(t)$ For every 0.5 msec

at Sampling proces $T_S = 0.5$ msec

$t = nT_S$

$2m = n \times 0.5m \qquad n = 4$

$n = 0 \dots N-1 \qquad N = 5$ Samples

$h(t) = -0.5t + 1$

$h(n) = -0.5nT_S + 1$

$h(n) = -0.25n + 1 \qquad n = 0 \dots 4$

$h(0) = 1$

$h(1) = 0.75 = 3/4$

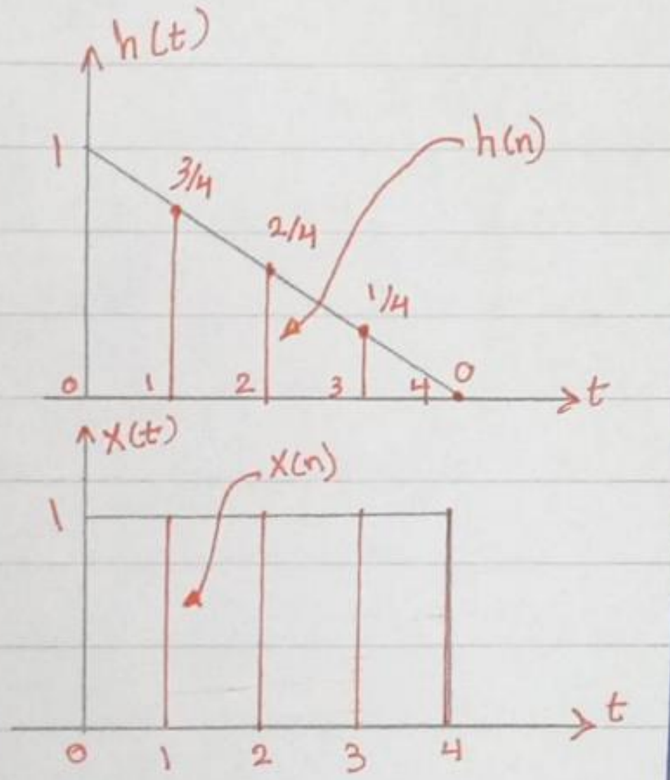
$h(2) = 2/4$

$h(3) = 1/4$

$h(4) = 0$

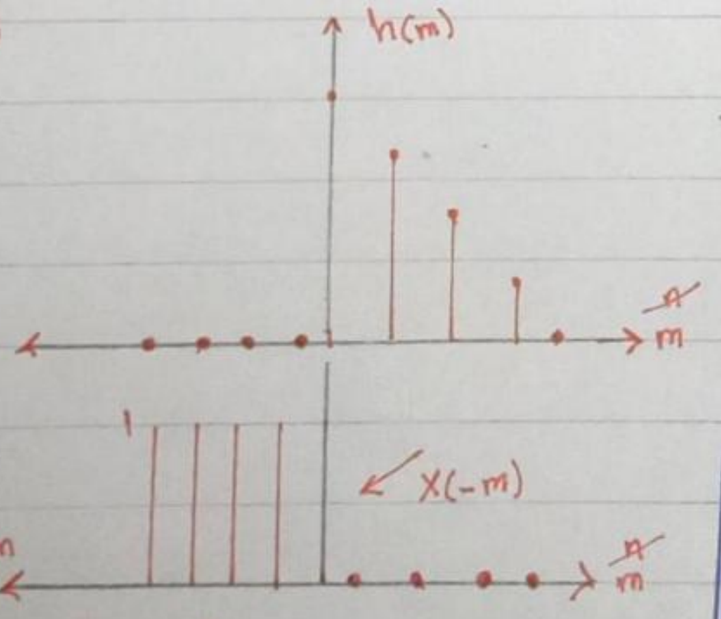
$h(n) = \{1, 3/4, 2/4, 1/4, 0\}$

$x(n) = \{1, 1, 1, 1, 1\}$



by Convolution Discrete_signal

$$y(n) = \sum_{m=0}^{N-1} h(m)\, x(n-m)$$

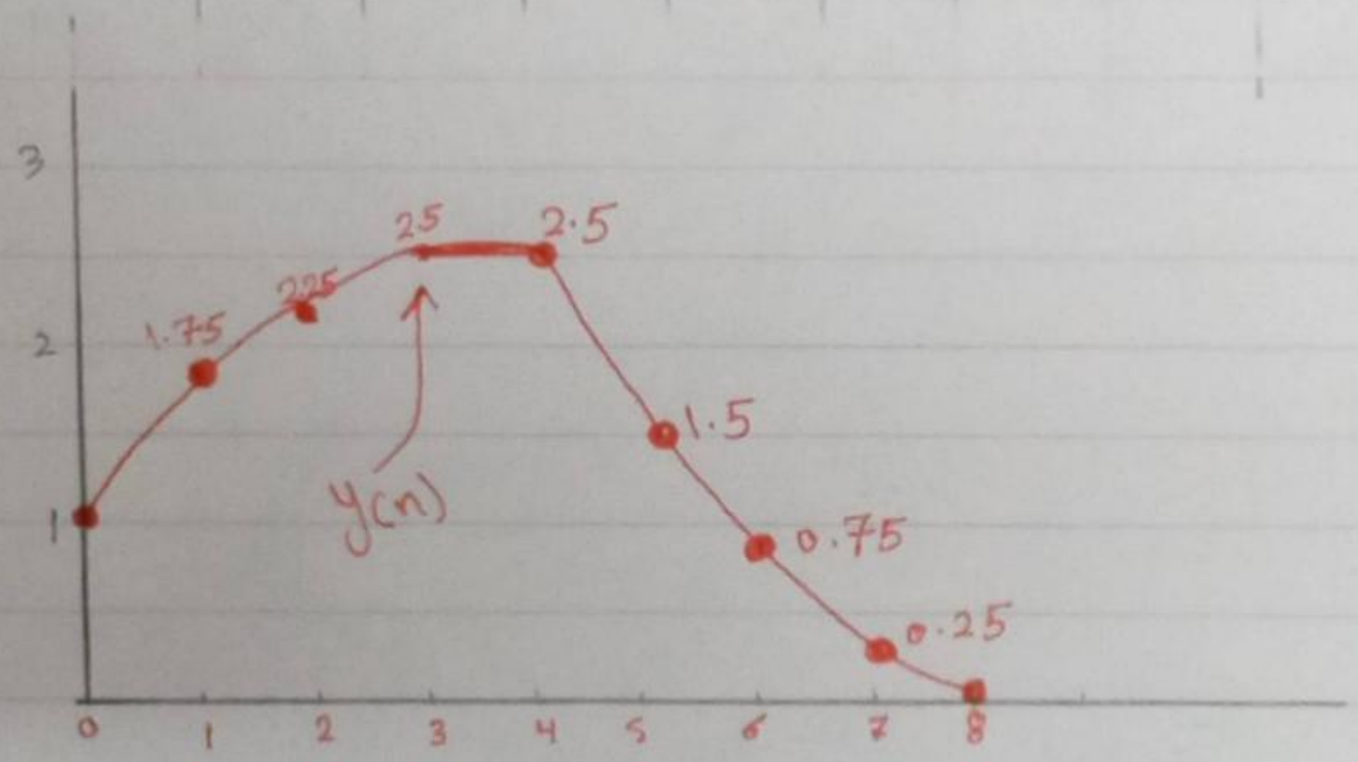


$h(m) = \{0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0.75 \quad 0.5 \quad 0.25 \quad 0\}$

$(-m) = \{1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0\}$

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n	h(m)	0	0	0	0	1	0.75	0.5	0.25	0	y(n)
0	x(0-m)	1	1	1	1	1	0	0	0	0	1
1	x(1-m)	0	1	1	1	1	1	0	0	0	1.75
2	x(2-m)	0	0	1	1	1	1	1	0	0	2.25
3	x(3-m)	0	0	0	1	1	1	1	1	0	2.5
4	x(4-m)	0	0	0	0	1	1	1	1	1	2.5
5	x(5-m)	0	0	0	0	0	0	1	1	1	1.5
6	x(6-m)	0	0	0	0	0	0	0	1	1	0.75
7	x(7-m)	0	0	0	0	0	0	0	0	1	0.25
8	x(8-m)	0	0	0	0	0	0	0	0	0	0



© Compare your results Part (a) and (B)

→ by comparing the waveform in Part (a) and (B) we can find the output obtainead in Fig (a) more accurate than (B).

→ To have y(n) closer to y(t), we should increase the Number of samples or by another word decrease the sampling interval.

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